
DIGITAL FORENSIC EXAMINATION REPORT

Analysis of a USB Flash Drive Image

CASE REFERENCE	USB-FD-2026-001
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EVIDENCE ITEM	flash.dd (Raw Forensic Image)
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EXAMINATION PLATFORM	SANS Investigative Forensic Toolkit (SIFT) Workstation
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TOOL SUITE	The Sleuth Kit (TSK)
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DATE OF REPORT	August 2, 2026
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1. Executive Summary

A digital forensic examination was conducted on a raw forensic image (`flash.dd`) acquired from a USB flash drive. The primary objectives were to verify image integrity, identify the underlying partition and file system structure, and analyze both active and deleted files. The examination was performed using industry-standard tools within the SIFT Workstation environment, ensuring the preservation of the original evidence's integrity.

2. Methodology

The investigation adhered to standard digital forensic procedures. The process involved:

- **Verification:** Confirmation of image format and technical metadata.
- **Structural Analysis:** Identification of partition tables and file system layouts.
- **Content Analysis:** Systematic listing and examination of stored data.
- **Recovery:** Extraction of specific artifacts and identification of deleted file remnants.

3. Examination Tasks & Findings

Task 1: Image Verification

Objective: Verify the source image technical specifications.

```
img_stat flash.dd
```

- **Image Type:** Raw (.dd)
- **Image Size:** 4,026,531,840 bytes
- **Sector Size:** 512 bytes

Analysis: The output confirms that flash.dd is a valid raw forensic image suitable for further technical examination.

Task 2: Partition Identification

Objective: Locate and identify the partition structure.

```
mmls flash.dd
```

- **Partition Table Type:** DOS (MBR)
- **Data Partition Start Sector:** 64

Analysis: The partition begins at sector 64, providing the necessary offset for file system-level commands.

Task 3: File System Analysis

Objective: Identify the file system and volume metadata.

```
fsstat -o 64 flash.dd
```

- **File System:** FAT32
- **Volume Label:** OTG ORG
- **Sector Size:** 512 bytes

Analysis: The image contains a FAT32 file system, commonly found on portable storage devices of this size.

Task 4: File Listing & Content Inventory

Objective: Enumerate all active files within the partition.

```
fls -o 64 flash.dd
```

- DIGITAL FORENSIC.pptx
- Group 1 Event log analysis.docx
- WriteBlockerSimulation.docx
- Chain_of_Custody_Practical_Enhanced.docx
- AnatomyOfMaliciousCodes-Analysis.pptx
- VMware-workstation-full-17.6.4-24832109.exe
- VMware-vdiskmanager.exe
- ts119a.exe
- DFRWS2005-RODEO.zip

Analysis: The inventory reveals a mix of forensic documentation, educational presentations, and executable installers.

Task 5: File Recovery

Objective: Recover a specific artifact for further analysis.

```
icat -o 64 flash.dd 52 > ts119a.exe
```

- **Target File:** ts119a.exe (Inode 52)

→ **Status:** Successfully recovered.

Analysis: The executable was successfully extracted from the forensic image without modification of the source material.

Task 6: Deleted File Identification

Objective: Identify files marked for deletion within the file system.

```
fls -r -d -o 64 flash.dd
```

→ _ankcard.zip

→ ~\$DIGITAL FORENSIC.pptx

→ (And other system/temp fragments)

Analysis: Several deleted files remain identifiable within the unallocated or slack space of the FAT32 file system, suggesting potential for deeper recovery operations.

4. Investigation Summary Table

Forensic Question	Technical Finding
Image Format Examined	Raw (.dd)
Partition Table Identified	DOS (MBR)
Partition Start Sector	64
Detected File System	FAT32
Volume Label	OTG ORG
Command to List Files	<code>fls -o 64 flash.dd</code>
Command to Recover File	<code>icat -o 64 flash.dd [inode] > [filename]</code>
Command to Identify Deleted Files	<code>fls -r -d -o 64 flash.dd</code>

5. Conclusion

The forensic examination of `flash.dd` successfully verified a valid raw image containing a **DOS (MBR)** partition starting at **sector 64**. The underlying **FAT32** file system, labeled **OTG ORG**, was thoroughly inventoried, revealing a variety of active files. Crucially, the identification of deleted entries confirms that data remains recoverable from the image. All procedures were conducted using **The Sleuth Kit**, ensuring forensic soundness and evidentiary integrity throughout the lifecycle of the examination.